CARVID 6.25 mg and 25 mg tablets
Carvedilol

FULL PRESCRIBING INFORMATION:

INDICATIONS AND USAGE:

Heart Failure: CARVID is indicated for the treatment of mild-to-severe chronic heart failure of ischemic or cardiomyopathic origin, usually in addition to diuretics, ACE inhibitors, and digitalis, to increase survival and, also, to reduce the risk of hospitalization.

Left Ventricular Dysfunction following Myocardial Infarction: Carvedilol is indicated to reduce cardiovascular mortality in clinically stable patients who have survived the acute phase of a myocardial infarction and have a left ventricular ejection fraction of less than or equal to 40% (with or without symptomatic heart failure).

Hypertension: CARVID is indicated for the management of essential hypertension. It can be used alone on in combination with other antihypertensive agents, especially thiazide-type diuretics.

DOSAGE AND ADMINISTRATION:

CARVID should be taken with food to slow the rate of absorption and reduce the incidence of orthostatic effects.

Heart Failure: DOSAGE MUST BE INDIVIDUALIZED AND CLOSELY MONITORED BY A PHYSICIAN DURING UP-TITRATION. Prior to initiation of CARVID, it is recommended that fluid retention be minimized. The recommended starting dose of CARVID is 3.125 mg twice daily for 2 weeks. If tolerated, patients may have their dose increased to 6.25, 12.5, and 25 mg twice daily over successive intervals of at least 2 weeks. Patients should be maintained on lower doses if higher doses are not tolerated. A maximum dose of 50 mg twice daily has been administered to patients with mild-to-moderate heart failure weighing over 85 kg (187 lbs). Patients should be advised that initiation of treatment and (to a lesser extent) dosage increases may be associated with transient symptoms of dizziness or lightheadedness (and rarely syncope) within the first hour after dosing. During these periods, patients should avoid situations such as driving or hazardous tasks, where symptoms could result in injury. Vasodilatory symptoms often do not require treatment, but it may be useful to separate the time of dosing of CARVID from that of the ACE inhibitor or to reduce temporarily the dose of the ACE inhibitor. The dose of CARVID should not be increased until symptoms of worsening heart failure or vasodilation have been stabilized. Fluid retention (with or without transient worsening heart failure symptoms) should be treated by an increase in the dose of diuretics. The dose of CARVID should be reduced if patients experience bradycardia (heart rate less than 55 beats per minute). Episodes of dizziness or fluid retention during initiation of CARVID can generally be Episodes of dizziness or fluid retention during initiation of CARVID can generally be managed without discontinuation of treatment and do not preclude subsequent successful titration of, or a favorable response to, carvedilol.

Left Ventricular Dysfunction following Myocardial Infarction: DOSAGE MUST BE INDIVIDUALIZED AND MONITORED DURING UP-TITRATION. Treatment with CARVID may be started as an inpatient or outpatient and should be started after the patient is hemodynamically stable and fluid retention has been minimized. It is recommended that CARVID be started at 6.25 mg twice daily and increased after 3 to 10 days, based on tolerability, to 12.5 mg twice daily, then again to the target dose of 25 mg twice daily. A lower starting dose may be used (3.125 mg twice daily) and/or the rate of up-titration may be starting dose may be used (3.125 mg twice daily) and/or the rate of up-titration may be slowed if clinically indicated (e.g., due to low blood pressure or heart rate, or fluid retention). Patients should be maintained on lower doses if higher doses are not tolerated. The recommended dosing regimen need not be altered in patients who received treatment with an IV or oral β- blocker during the acute phase of the myocardial infarction.

Hypertension: DOSAGE MUST BE INDIVIDUALIZED. The recommended starting dose of CARVID is 6.25 mg twice daily. If this dose is tolerated, using standing systolic pressure measured about1 hour after dosing as a guide, the dose should be maintained for 7 to 14 days, and then increased to 12.5 mg twice daily if needed, based on trough blood pressure, again using standing systolic pressure 1 hour after dosing as a guide for tolerance. This dose should also be maintained for 7 to14 days and can then be adjusted upward to 25 mg twice daily if tolerated and needed. The full antihypertensive effect of CARVID is seen within 7 to 14 days. Total daily dose should not exceed 50 mg. Concomitant administration with a diuretic can be expected to produce additive effects and exaggerate the orthostatic component of carvedilol action.

Hepatic Impairment: CARVID should not be given to patients with severe hepatic impairment.

DOSAGE FORMS AND STRENGTHS:

White, round biconvex, tablet scored from one side and engraved with MAP from the other side. Carvid 6.25 mg. Carvid 25 mg.

COMPOSITION:

Active Ingredient: Carvedilol.

Inactive Ingredients: Microcrystalline cellulose (Avicel pH 101), Lactose fast flow, Croscarmellose sodium (Acidsol), Magnesium Stearate, Polyvinylpyrrolidone (Povidone K 25).

CONTRAINDICATIONS:

CARVID is contraindicated in the following conditions: Bronchial asthma or related bronchospastic conditions. Deaths from status asthmatics have been reported following single doses of CARVID. Second- or third-degree AV block. Sick sinus syndrome. Severe bradycardia (unless a permanent pacemaker is in place). Patients with cardiogenic shock or who have decompensated heart failure requiring the use of intravenous inotropic therapy. Such patients should first be weaned for intravenous therapy before initiating CARVID. Patients with severe hepatic impairment. Patients with a history of a serious hypersensitivity reaction (e.g., Stevens-Johnson syndrome, anaphylactic reaction, angioedema) to any component of this medication or other medications containing carvedilol.

WARNINGS AND PRECAUTIONS:

Cessation of Therapy: Patients with coronary artery disease, who are being treated with CARVID, should be advised against abrupt discontinuation of therapy. Severe exacerbation of angina and the occurrence of myocardial infarction and ventricular arrhythmias have been reported in angina patients following the abrupt discontinuation of therapy with β-blockers. The last 2 complications may occur with or without preceding exacerbation of the angina pectoris. As with other β-blockers, when discontinuation of CARVID is planned, the patients should be carefully observed and advised to limit physical activity to a minimum. CARVID should be discontinued over 1 to 2 weeks whenever possible. If the angina worsens or acute coronary insufficiency develops, it is recommended that CARVID be promptly reinstituted, at least temporarily. Because coronary artery disease is common and may be unrecognized, it may be prudent not to discontinue therapy with CARVID abruptly even in patients treated only for hypertension or heart failure.

Bradycardia: In clinical trials, Carvedilol caused bradycardia in about 2% of hypertensive subjects, 9% of heart failure subjects, and 6.5% of myocardial infarction subjects with left ventricular dysfunction. If pulse rate drops below 55 beats per minute, the dosage should be reduced.

Hypotension: In clinical trials of primarily mild-to-moderate heart failure, hypotension and postural Hypotension. Starting with a low dose, administration with food, and gradual up-titration should decrease the likelihood of syncope or excessive hypotension. During initiation of therapy, the patient should be cautioned to avoid situations such as driving or hazardous tasks, where injury could result should syncope occur.

Heart Failure/Fluid Retention: Worsening heart failure or fluid retention may occur during up-titration of carvedilol. If such symptoms occur, diuretics should be increased and the carvedilol dose should not be advanced until clinical stability resumes. Occasionally it is necessary to lower the carvedilol dose or temporarily discontinue it. Such episodes do not preclude subsequent successful titration of, or a favorable response to, carvedilol. In a placebo-controlled trial of subjects with severe heart failure, worsening heart failure during the first 3 months was reported to a similar degree with carvedilol and with placebo. When treatment was maintained beyond 3 months, worsening heart failure was reported less frequently in subjects treated with carvedilol than with placebo. Worsening heart failure observed during long-term therapy is more likely to be related to the patients' underlying disease than to treatment with carvedilol.

Non-allergic Bronchospasm: Patients with bronchospastic disease (e.g., chronic bronchitis and emphysema) should, in general, not receive β-blockers. CARVID may be used with caution, however, in patients who do not respond to, or cannot tolerate, other antihypertensive agents. It is prudent, if CARVID is used, to use the smallest effective dose, so that inhibition of endogenous or exogenous β-agonists is minimized. In clinical trials of subjects with heart failure, subjects with bronchospastic disease were enrolled if they did not require oral or inhaled medication to treat their bronchospastic disease. In such patients, it is recommended that carvedilol be used with caution. The dosing recommendations should be followed closely and the dose should be lowered if any evidence of bronchospasm is observed during up-titration.

Glycemic Control in Type 2 Diabetes: In general, β-blockers may mask some of the manifestations of hypoglycemia, particularly tachycardia. Nonselective β-blockers may potentiate insulin-induced hypoglycemia and delay recovery of serum glucose levels. Patients subject to spontaneous hypoglycemia, or diabetic patients receiving insulin or oral hypoglycemic agents, should be cautioned about these possibilities. In heart failure patients with diabetes, carvedilol therapy may lead to worsening hyperglycemia, which responds to intensification of hypoglycemic therapy. It is recommended that blood glucose be monitored when carvedilol dosing is initiated, adjusted, or discontinued. Trials designed to examine the effects of carvedilol on glycemic control in patients with diabetes and heart failure have not been conducted. In a trial designed to examine the effects of carvedilol on glycemic control in a population with mild-to-moderate hypertension and well-controlled type 2 diabetes mellitus, carvedilol had no adverse effect on glycemic control, based on HbA1c measurements.

Peripheral Vascular Disease: β-blockers can precipitate or aggravate symptoms of arterial insufficiency in patients with peripheral vascular disease. Caution should be exercised in such individuals.

Deterioration of Renal Function: Rarely, use of carvedilol in patients with heart failure has resulted in deterioration of renal function. Patients at risk appear to be those with low blood pressure (systolic blood pressure less than100 mm Hg), ischemic heart disease and diffuse vascular disease, and/or underlying renal insufficiency. Renal function has returned to baseline when carvedilol was stopped. In patients with these risk factors it is recommended that renal function be monitored during up-titration of carvedilol and the drug discontinued or dosage reduced if worsening of renal function occurs.

Major Surgery: Chronically administered beta-blocking therapy should not be routinely withdrawn prior to major surgery; however, the impaired ability of the heart to respond to reflex adrenergic stimuli may augment the risks of general anesthesia and surgical procedures.

Thyrotoxicosis: β-adrenergic blockade may mask clinical signs of hyperthyroidism, such as tachycardia. Abrupt withdrawal of β-blockade may be followed by an exacerbation of the symptoms of hyperthyroidism or may precipitate thyroid storm.

Pheochromocytoma: In patients with pheochromocytoma, an α-blocking agent should be initiated prior to the use of any β-blocking agent. Although carvedilol has both α- and β-blocking pharmacologic activities, there has been no experience with its use in this condition. Therefore, caution should be taken in the administration of carvedilol to patients suspected of having pheochromocytoma.

Prinzmetal's Variant Angina: Agents with non-selective β-blocking activity may provoke chest pain in patients with Prinzmetal's variant angina. There has been no clinical experience with carvedilol in these patients although the α-blocking activity may prevent such symptoms. However, caution should be taken in the administration of carvedilol to patients suspected of having Prinzmetal's variant angina.

Risk of Anaphylactic Reaction: While taking β-blockers, patients with a history of severe anaphylactic reaction to a variety of allergens may be more reactive to repeated challenge, either accidental, diagnostic, or therapeutic. Such patients may be unresponsive to the usual doses of epinephrine used to treat allergic reaction.

Intraoperative Floppy Iris Syndrome: Intraoperative Floppy Iris Syndrome (IFIS) has been observed during cataract surgery in some patients treated with alpha-1 blockers (CARVID is an alpha/beta blocker). This variant of small pupil syndrome is characterized by the combination of a flaccid iris that billows in response to intraoperative irrigation currents, progressive intraoperative miosis despite preoperative dilation with standard mydriatic drugs, and potential prolapse of the iris toward the phacoemulsification incisions. The patient's ophthalmologist should be prepared for possible modifications to the surgical technique, such as utilization of iris hooks, iris dilator rings, or viscoelastic substances. There does not appear to be a benefit of stopping alpha-1 blocker therapy prior to cataract surgery.

ADVERSE REACTIONS:

Clinical Studies Experience: CARVID has been evaluated for safety in subjects with heart failure (mild, moderate, and severe), in subjects with left ventricular dysfunction following myocardial infarction and in hypertensive subjects. The observed adverse event profile was consistent with the pharmacology of the drug and the health status of the subjects in the clinical trials. Adverse events reported for each of these patient populations are provided below. Excluded are adverse events considered too general to be informative, and those not reasonably associated with the use of the drug because they were associated with the condition being treated or are very common in the treated population. Rates of adverse events were generally similar across demographic subsets (men and women, elderly and non-elderly, blacks and non-blacks).

Heart Failure: Adverse events reported in subjects with mild-to-moderate heart failure enrolled in US placebo-controlled clinical trials, and with severe heart failure enrolled in the COPERNICUS trial. Shown are adverse events that occurred more frequently in drug-treated subjects than placebo-treated subjects with an incidence of greater than 3% in subjects treated with carvedilol regardless of causality. Median trial medication exposure was 6.3 months for both carvedilol and placebo subjects in the trials of mild-to-moderate heart failure, and 10.4 months in the trial of severe heart failure subjects. The adverse event profile of CARVID observed in the long-term COMET trial was generally similar to that observed in the US Heart Failure Trials.

Body as a Whole: Asthenia Fatigue. Digoxin level increased. Edema generalized. Edema dependent.

Cardiovascular: Bradycardia Hypotension Syncope. Angina pectoris.

Central Nervous System: Dizziness. Headache.

Gastrointestinal: Diarrhea Nausea. Vomiting.

Metabolic: Hyperglycemia Weight increase BUN increased NPN increased Hypercholesterolemia. Edema peripheral.

Musculoskeletal: Arthralgia.

Respiratory: Cough increased Rales.

Vision: Vision abnormal. Cardiac failure and dyspnea were also reported in these trials, but the rates were equal or greater in subjects who received placebo.

Incidence greater than 1% to less than or equal to 3%: Body as a Whole: Allergy, malaise, hypovolemia, fever, leg edema. Cardiovascular: Fluid overload, postural hypotension, aggravated angina pectoris, AV block, palpitation, hypertension. Central and Peripheral Nervous System: Hypesthesia, vertigo, paresthesia. Gastrointestinal: Melena, periodontitis. Liver and Biliary System: SGPT increased, SGOT increased. Metabolic and Nutritional: Hyperuricemia, hypoglycemia, hyponatremia, increased alkaline phosphatase, glycosuria, hypervolemia, diabetes mellitus, GGT increased, weight loss, hyperkalemia, creatinine increased. Musculoskeletal: Muscle cramps. Platelet, Bleeding, and Clotting: Prothrombin decreased, purpura, thrombocytopenia. Psychiatric: Somnolence. Reproductive, male: Impotence. Special Senses: Blurred vision. Urinary System: Renal insufficiency, albuminuria, hematuria.

Left Ventricular Dysfunction following Myocardial Infarction: Carvedilol has been evaluated for safety in survivors of an acute myocardial infarction with left ventricular dysfunction .The most common adverse events reported with CARVID in the CAPRICORN trial were consistent with the profile of the drug in the US heart failure trials and the COPERNICUS trial. The only additional adverse events reported in CAPRICORN in greater than 3% of the subjects and more commonly on carvedilol were dyspnea, anemia, and lung edema. The following adverse events were reported with a frequency of greater than 1% but less than or equal to 3% and more frequently with CARVID : flu syndrome, cerebrovascular accident, peripheral vascular disorder, hypotonia, depression, gastrointestinal pain, arthritis, and gout. The overall rates of discontinuations due to adverse events were similar in both groups of subjects. In this database, the only cause of discontinuation greater than 1%, and occurring more often on carvedilol was hypotension (1.5% on carvedilol, 0.2% on placebo). Adverse events in US placebo-controlled clinical trials for hypertension that occurred with an incidence of greater than or equal to 1% regardless of causality, and that were more frequent in drug-treated subjects than placebo-treated subjects.

BODY SYSTEM / ADVERSE EVENT:

Cardiovascular: Bradycardia. Postural hypotension. Peripheral edema.

Central Nervous System: Dizziness. Insomnia.

Gastrointestinal: Diarrhea.

Hematologic: Thrombocytopenia.

Metabolic: Hypertriglyceridemia. Dyspnea and fatigue were also reported in these trials, but the rates were equal or greater in subjects who received placebo. The following adverse events not described above were reported as possibly or probably related to CARVID in worldwide open or controlled trials with CARVID in subjects with hypertension or heart failure.

Incidence greater than 0.1% to less than or equal to 1%:

Cardiovascular: Peripheral ischemia, tachycardia.

Central and Peripheral Nervous System: Hypokinesia.

Gastrointestinal: Bilirubinemia, increased hepatic enzymes (0.2% of hypertension patients and 0.4% of heart failure patients were discontinued from therapy because of increases in hepatic enzymes).

Psychiatric: Nervousness, sleep disorder, aggravated depression, impaired concentration, abnormal thinking, paroniria, emotional lability.

Respiratory System: Asthma.

Reproductive, male: Decreased libido.

Skin and Appendages: Pruritus, rash erythematous, rash maculopapular, rash psoriaform, photosensitivity reaction.

Special Senses: Tinnitus.

Urinary System: Micturition frequency increased.

Autonomic Nervous System: Dry mouth, sweating increased.

Metabolic and Nutritional: Hypokalemia, hypertriglyceridemia.

Hematologic: Anemia, leukopenia.The following events were reported in less than or equal to 0.1% of subjects and are potentially important: complete AV block, bundle branch block, myocardial ischemia, cerebrovascular disorder, convulsions, migraine, neuralgia, paresis, anaphylactoid reaction, alopecia, exfoliative dermatitis, amnesia, GI hemorrhage, bronchospasm, pulmonary edema, decreased hearing, respiratory alkalosis, increased BUN, decreased HDL, pancytopenia, and atypical lymphocytes.

Laboratory Abnormalities: Reversible elevations in serum transaminases (ALT or AST) have been observed during treatment with CARVID. Rates of transaminase elevations (2 to 3 times the upper limit of normal) observed during controlled clinical trials have generally been similar between subjects treated with CARVID and those treated with placebo. However, transaminase elevations, confirmed by rechallenge, have been observed with CARVID. In a long-term, placebo-controlled trial in severe heart failure, subjects treated with CARVID had lower values for hepatic transaminases than subjects treated with placebo, possibly because Improvements in cardiac function induced by CARVID led to less hepatic congestion and/or improved hepatic blood flow. CARVID has not been associated with clinically significant changes in serum potassium, total triglycerides, total cholesterol, HDL cholesterol, uric acid, blood urea nitrogen, or creatinine. No clinically relevant changes were noted in fasting serum glucose in hypertensive patients; fasting serum glucose was not evaluated in the heart failure clinical trials.

Postmarketing Experience: The following adverse reactions have been identified during post-approval use of Carvedilol. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

Blood and Lymphatic System Disorders: Aplastic anemia.

Immune System Disorders: Hypersensitivity (e.g., anaphylactic reactions, angioedema, urticaria).

Renal and Urinary Disorders: Urinary incontinence.

Respiratory, Thoracic, and Mediastinal Disorders: Interstitial pneumonitis.

Skin and Subcutaneous Tissue Disorders: Stevens-Johnson syndrome, toxic epidermal necrolysis, erythema multiforme.

DRUG INTERACTIONS:

CYP2D6 Inhibitors and Poor Metabolizers: Interactions of carvedilol with potent inhibitors of CYP2D6 isoenzyme (such as quinidine, fluoxetine, paroxetine, and propafenone) have not been studied, but these drugs would be expected to increase blood levels of the R(+) enantiomer of carvedilol. Retrospective analysis of side effects in clinical trials showed that poor 2D6 metabolizers had a higher rate of dizziness during up-titration, presumably resulting from vasodilating effects of the higher concentrations of the α-blocking R(+) enantiomer.

Hypotensive Agents: Patients taking both agents with β-blocking properties and a drug that can deplete catecholamines (e.g., reserpine and monoamine oxidase inhibitors) should be observed closely for signs of hypotension and/or severe bradycardia. Concomitant administration of clonidine with agents with β-blocking properties may potentiate blood-pressure- and heart-rate-lowering effects. When concomitant treatment with agents with β-blocking properties and clonidine is to be terminated, the β-blocking agent should be discontinued first. Clonidine therapy can then be discontinued several days later by gradually decreasing the dosage.

Cyclosporine: Modest increases in mean trough cyclosporine concentrations were observed following initiation of carvedilol treatment in 21 renal transplant subjects suffering from chronic vascular rejection. In about 30% of subjects, the dose of cyclosporine had to be reduced in order to maintain cyclosporine concentrations within the therapeutic range, while in the remainder no adjustment was needed. On the average for the group, the dose of cyclosporine was reduced about 20% in these subjects. Due to wide interindividual variability in the dose adjustment required, it is recommended that cyclosporine concentrations be monitored closely after initiation of carvedilol therapy and that the dose of cyclosporine be adjusted as appropriate.

Digitalis Glycosides: Both digitalis glycosides and β-blockers slow atrioventricular conduction and decrease heart rate. Concomitant use can increase the risk of bradycardia. Digoxin concentrations are increased by about 15% when digoxin and carvedilol are administered concomitantly. Therefore, increased monitoring of digoxin is recommended when initiating, adjusting, or discontinuing CARVID .

Inducers/Inhibitors of Hepatic Metabolism: Rifampin reduced plasma concentrations of carvedilol by about 70% Cimetidine increased AUC by about 30% but caused no change in C_{max}.

Amiodarone: Amiodarone, and its metabolite desethyl amiodarone, inhibitors of CYP2C9, and P-glycoprotein increased concentrations of the S(-)-enantiomer of carvedilol by at least 2 fold.The concomitant administration of amiodarone or other CYP2C9 inhibitors such as fluconazole with CARVID may enhance the β-blocking properties of carvedilol resulting in further slowing of the heart rate or cardiac conduction. Patients should be observed for signs of bradycardia or heart block, particularly when one agent is added to pre-existing treatment with the other.

Calcium Channel Blockers: Conduction disturbance (rarely with hemodynamic compromise) has been observed when CARVID is coadministered with diltiazem. As with other agents with β-blocking properties, if CARVID is to be administered with calcium channel blockers of the verapamil or diltiazem type, it is recommended that ECG and blood pressure be monitored.

Insulin or Oral Hypoglycemics: Agents with β-blocking properties may enhance the blood-sugar-reducing effect of insulin and oral hypoglycemics. Therefore, in patients taking insulin or oral hypoglycemics, regular monitoring of blood glucose is recommended.

Anesthesia: If treatment with CARVID is to be continued perioperatively, particular care should be taken when anesthetic agents that depress myocardial function, such as ether, cyclopropane, and trichloroethylene, are used.

USE IN SPECIFIC POPULATIONS:

Pregnancy: Pregnancy Category C. There are no adequate and well-controlled studies in pregnant women. CARVID should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Nursing Mothers: It is not known whether this drug is excreted in human milk.Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants from β-blockers, especially bradycardia, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother. The effects of other α- and β-blocking agents have included perinatal and neonatal distress.

Geriatric Use: With the exception of dizziness in hypertensive subjects no overall differences in the safety or effectiveness were observed between the older subjects and younger subjects in each of these populations. Similarly, other reported clinical experience has not identified differences in responses between the elderly and younger subjects, but greater sensitivity of some older individuals cannot be ruled out.

OVERDOSAGE:

Over dosage may cause severe hypotension, bradycardia, cardiac insufficiency, cardiogenic shock, and cardiac arrest. Respiratory problems, bronchospasms, vomiting, lapses of consciousness, and generalized seizures may also occur. The patient should be placed in a supine position and, where necessary, kept under observation and treated under intensive-care conditions.

The following agents may be administered: For excessive bradycardia: Atropine, 2 mg IV. To support cardiovascular function: Glucagon, 5 to 10 mg IV rapidly over 30 seconds, followed by a continuous infusion of 5 mg per hour; sympathomimetics (dobutamine, isoprenaline, adrenaline) at doses according to body weight and effect. If peripheral vasodilation dominates, it may be necessary to administer adrenaline or noradrenaline with continuous monitoring of circulatory conditions. For therapy-resistant bradycardia, pacemaker therapy should be performed. For bronchospasm, β-sympathomimetics (as aerosol or IV) or aminophylline IV should be given. In the event of seizures, slow IV injection of diazepam or clonazepam is recommended.

NOTE: In the event of severe intoxication where there are symptoms of shock, treatment with antidotes must be continued for a sufficiently long period of time consistent with the 7- to 10-hour half-life of carvedilol.

CLINICAL PHARMACOLOGY:

Mechanism of Action: CARVID is a racemic mixture in which nonselective β-adrenoreceptor blocking activity is present in the S(-) enantiomer and α1-adrenergic blocking activity is present in both R(+) and S(-) enantiomers at equal potency. CARVID has no intrinsic sympathomimetic activity.

PHARMACODYNAMICS:

Heart Failure: The basis for the beneficial effects of CARVID in heart failure is not established.

Left Ventricular Dysfunction following Myocardial Infarction: The basis for the beneficial effects of CARVID in patients with left ventricular dysfunction following an acute myocardial infarction is not established.

Hypertension: The mechanism by which β-blockade produces an antihypertensive effect has not been established. β-adrenoreceptor blocking activity has been demonstrated in animal and human studies showing that carvedilol (1) reduces cardiac output in normal subjects; (2) reduces exercise- and/or isoproterenol-induced tachycardia; and (3) reduces reflex orthostatic tachycardia. Significant β-adrenoreceptor blocking effect is usually seen within 1 hour of drug administration. α1-adrenoreceptor blocking activity has been demonstrated in human and animal studies, showing that carvedilol (1) attenuates the pressor effects of phenylephrine; (2) causes vasodilation; and (3) reduces peripheral vascular resistance. These effects contribute to the reduction of blood pressure and usually are seen within 30 minutes of drug administration. Due to the α1-receptor blocking activity of carvedilol, blood pressure is lowered more in the standing than in the supine position, and symptoms of postural hypotension (1.8%), including rare instances of syncope, can occur. Following oral administration, when postural hypotension has occurred, it has been transient and is uncommon when CARVID is administered with food at the recommended starting dose and titration increments are closely followed In hypertensive patients with normal renal function, therapeutic doses of CARVID decreased renal vascular resistance with no change in glomerular filtration rate or renal plasma flow. Changes in excretion of sodium, potassium, uric acid, and phosphorus in hypertensive patients with normal renal function were similar after CARVID and placebo. CARVID has little effect on plasma catecholamines, plasma aldosterone, or electrolyte levels, but it does significantly reduce plasma renin activity when given for at least 4 weeks. It also increases levels of atrial natriuretic peptide.

PHARMACOKINETICS:

CARVID is rapidly and extensively absorbed following oral administration, with absolute bioavailability of approximately 25% to 35% due to a significant degree of first-pass metabolism. Following oral administration, the apparent mean terminal elimination half-life of carvedilol generally ranges from 7 to 10 hours. Plasma concentrations achieved are proportional to the oral dose administered. When administered with food, the rate of absorption is slowed, as evidenced by a delay in the time to reach peak plasma levels, with no significant difference in extent of bioavailability. Taking CARVID with food should minimize the risk of orthostatic hypotension. Carvedilol is extensively metabolized. Following oral administration of radiolabelled carvedilol to healthy volunteers, carvedilol accounted for only about 7% of the total radioactivity in plasma as measured by area under the curve (AUC). Less than 2% of the dose was excreted unchanged in the urine. Carvedilol is metabolized primarily by aromatic ring oxidation and glucuronidation. The oxidative metabolites are further metabolized by conjugation via glucuronidation and sulfation. The metabolites of carvedilol are excreted primarily via the bile into the feces. Demethylation and hydroxylation at the phenol ring produce 3 active metabolites with β-receptor blocking activity. Based on preclinical studies, the 4'-hydroxyphenyl metabolite is approximately 13 times more potent than carvedilol for β-blockade. Compared with carvedilol, the 3 active metabolites exhibit weak vasodilating activity. Plasma concentrations of the active metabolites are about one-tenth of those observed for carvedilol and have pharmacokinetics similar to the parent. Carvedilol undergoes stereoselective first-pass metabolism with plasma levels of R(+)-carvedilol approximately 2 to 3 times higher than S(-)-carvedilol following oral administration in healthy subjects. The mean apparent terminal elimination half-lives for R(+)-carvedilol range from 5 to 9 hours compared with 7 to 11 hours for the S(-)-enantiomer.The primary P450 enzymes responsible for the metabolism of both R(+) and S(-)-carvedilol in human liver microsomes were CYP2D6 and CYP2C9 and to a lesser extent CYP3A4, 2C19,1A2, and 2E1. CYP2D6 is thought to be the major enzyme in the 4'- and 5'-hydroxylation of carvedilol, with a potential contribution from 3A4. CYP2C9 is thought to be of primary importance in the O-methylation pathway of S(-carvedilol. Carvedilol is subject to the effects of genetic polymorphism with poor metabolizers of debrisoquin (a marker for cytochrome P450 2D6) exhibiting 2- to 3-fold higher plasma concentrations of R(+)-carvedilol compared with extensive metabolizers. In contrast, plasma levels of S(-)-carvedilol are increased only about 20% to 25% in poor metabolizers, indicating this enantiomer is metabolized to a lesser extent by cytochrome P450 2D6 than R(+)-carvedilol. The pharmacokinetics of carvedilol do not appear to be different in poor metabolizers of S-mephenytoin (patients deficient in cytochrome P450 2C19). Carvedilol is more than 98% bound to plasma proteins, primarily with albumin. The plasma-protein binding is independent of concentration over the therapeutic range. Carvedilol is a basic, lipophilic compound with a steady-state volume of distribution of approximately 115 L, indicating substantial distribution into extravascular tissues. Plasma clearance ranges from 500 to 700 mL/min.

Specific Populations: Heart Failure: Steady-state plasma concentrations of carvedilol and its enantiomers increased proportionally over the 6.25- to 50- mg dose range in subjects with heart failure. Compared with healthy subjects, heart failure subjects had increased mean AUC and C_{max} values for carvedilol and its enantiomers, with up to 50% to 100% higher values observed in 6 subjects with NYHA class IV heart failure. The mean apparent terminal elimination half-life for carvedilol was similar to that observed in healthy subjects.

Geriatric: Plasma levels of carvedilol average about 50% higher in the elderly compared with young subjects.

Hepatic Impairment: Compared with healthy subjects, patients with severe liver impairment (cirrhosis) exhibit a 4- to 7-fold increase in carvedilol levels. Carvedilol is contraindicated in patients with severe liver impairment.

Renal Impairment: Although carvedilol is metabolized primarily by the liver, plasma concentrations of carvedilol have been reported to be increased in patients with renal impairment. Based on mean AUC data, approximately 40% to 50% higher plasma concentrations of carvedilol were observed in hypertensive subjects with moderate to severe renal impairment compared with a control group of hypertensive subjects with normal renal function. However, the ranges of AUC values were similar for both groups. Changes in mean peak plasma levels were less pronounced, approximately 12% to 26% higher in subjects with impaired renal function. Consistent with its high degree of plasma protein-binding, carvedilol does not appear to be cleared significantly by hemodialysis.

DRUG-DRUG INTERACTIONS:

Since carvedilol undergoes substantial oxidative metabolism, the metabolism and pharmacokinetics of carvedilol may be affected by induction or inhibition of cytochrome P450 enzymes.

Amiodarone: In a pharmacokinetic trial conducted in 106 Japanese subjects with heart failure, coadministration of small loading and maintenance doses of amiodarone with carvedilol resulted in at least a 2-fold increase in the steady-state trough concentrations of S(-)-carvedilol.

Cimetidine: In a pharmacokinetic trial conducted in 10 healthy male subjects, cimetidine (1,000 mg per day) increased the steady-state AUC of carvedilol by 30% with no change in C_{max}.

Digoxin: Following concomitant administration of carvedilol (25 mg once daily) and digoxin (0.25 mg once daily) for 14 days, steady-state AUC and trough concentrations of digoxin were increased by 14% and 16%, respectively, in 12 hypertensive subjects.

Glyburide: In 12 healthy subjects, combined administration of carvedilol (25 mg once daily) and a single dose of glyburide did not result in a clinically relevant pharmacokinetic interaction for either compound.

Hydrochlorothiazide: A single oral dose of carvedilol 25 mg did not alter the pharmacokinetics of a single oral dose of hydrochlorothiazide 25 mg in 12 subjects with hypertension. Likewise, hydrochlorothiazide had no effect on the pharmacokinetics of carvedilol.

Rifampin: In a pharmacokinetic trial conducted in 8 healthy male subjects, rifampin (600 mg daily for 12 days) decreased the AUC and C_{max} of carvedilol by about 70% [see Drug Interactions (7.5)].

Torsemide: In a trial of 12 healthy subjects, combined oral administration of carvedilol 25 mg once daily and torsemide 5 mg once daily for 5 days did not result in any significant differences in their pharmacokinetics compared with administration of the drugs alone.

Warfarin: Carvedilol (12.5 mg twice daily) did not have an effect on the steady-state prothrombin time ratios and did not alter the pharmacokinetics of R(+)- and S(-)-warfarin following concomitant administration with warfarin in 9 healthy volunteers.

Hypertension: In general, as is true for other β-blockers, responses were smaller in black than non-black subjects. There were no age- or gender-related differences in response. The peak antihypertensive effect occurred 1 to 2 hours after a dose. The dose-related blood pressure response was accompanied by a dose-related increase in adverse effects.

Hypertension with Type 2 Diabetes Mellitus: CARVID had no adverse effect on glycemic control, based on HbA1c measurements (mean change from baseline of 0.02%, 95% CI: -0.06 to 0.10, P = NS).

HOW SUPPLIED/STORAGE AND HANDLING:

- Both 6.25 mg and 25 mg are White, round biconvex, tablet scored from one side and engraved with MAP from the other side.
- Stored in a temperature not exceeding 30°C in a dry place.
- Pack of 6.25 mg is a carton box containing (Al / white opaque PVC) blister of 10 tablets + insert leaflet.
- Pack of 25 mg is a carton box containing (Al / white opaque PVC) strips each of 10 tablets + insert leaflet.

Patients taking CARVID should be advised of the following:

- Patients should take CARVID with food.
- Patients should not interrupt or discontinue using CARVID without a physician's advice.
- Patients with heart failure should consult their physician if they experience signs or symptoms of worsening heart failure such as weight gain or increasing shortness of breath.
- Patients may experience a drop in blood pressure when standing, resulting in dizziness and, rarely, fainting. Patients should sit or lie down when these symptoms of lowered pressure occur.
- If experiencing dizziness or fatigue, patients should avoid driving or hazardous tasks.
- Patients should consult a physician if they experience dizziness or faintness, in case the dosage should be adjusted.
- Diabetic patients should report any changes in blood sugar levels to their physician.
- Contact lens wearers may experience decreased lacrimation.

Keep All Medicines Out of Reach of Children

Manufactured by: Apex Pharma - S.A.E. - Badr City - Egypt.

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